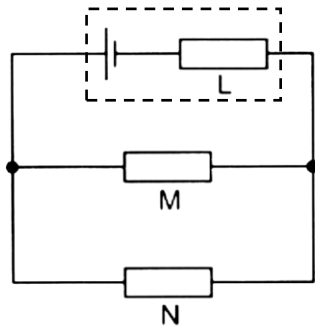


$$E = DV = 1.5(2.0 \times 10^{-3}) = 3.0 \times 10^{-3} \text{ N C}$$

- 22** A battery supplies a current of 0.025 A for 80 s. During this time it produces 18.0 J of electrical energy while resistor M receives 11.0 J and resistor N receives 4.0 J.



What is the e.m.f. of the battery and its internal resistance L ?

	e.m.f. of battery / V	internal resistance L of battery / Ω
A	1.5	60
B	9.0	60
C	9.0	360
D	16.5	360

Ans: B

$$E_{\text{batt}} = VIt$$

$$18.0 = \mathcal{E}(0.025)(80)$$

$$\mathcal{E} = 9.0 \text{ V}$$

$$E_r = I^2 rt$$

$$18.0 - 11.0 - 4.0 = 0.025^2 r(80)$$

$$r = 60 \Omega$$